



Genome-wide analysis of the AP2/ERF gene family in *Populus trichocarpa*

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ABSTRACT

Populus is a model system for investigating the wood development, crown formation, and disease resistance in perennial plants. AR2/ERF is a large family of transcription factors in plant, encoding transcriptional regulators with a variety of functions involved in the developmental and physiological processes. Here, starting from database of *Populus* genome, we identified 200 AP2/ERF genes by *in silico* cloning method using the AP2/ERF conserved domain amino acid sequence of *Arabidopsis thaliana* as probe. Based on the number of AP2/ERF domains and the function of the genes, those AP2/ERF genes from *Populus* were classified into four subfamilies named the AP2, DREB, ERF, RAV, and a soloist. Among these genes, the number genes of total AP2/ERF family genes, DREB subfamily, and ERF subfamily from *Populus trichocarpa* were about 1.4–1.6-fold than those from *A. thaliana*. The rates were very similar for the putative homologs between *Populus* and *Arabidopsis*.

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The genus *Populus*, such as poplar, aspen, and cottonwoods, all of which are found in the Northern hemisphere, makes it a valuable forest resource and an important ecological species [1,2]. *Populus* have been used all over the world to produce a large variety of paper-making and timber products. It is an indispensable component of paper, plywood, and many other wood products. This wood species is also a suitable model for plant biology, such as using as a model to study wood formation [3]. The advantages of using *Populus* as a model species were its rapid growth process, simple propagation method, and easy genetic transformation.

Plant growth and productivity are greatly affected by environmental stresses such as drought, high salinity, and low temperature, which have an adverse effect on the growth of plants and the productivity of crops. Expression of a variety of genes is induced by these stresses in various plants. The products of these genes function not only in stress tolerance but also in the regulation of gene expression and signal transduction in stress responses [4]. Genome-wide analyses have identified thousands of genes encoding transcription factors that are induced or repressed by a range of environmental stresses. Various transcription factors and *cis*-acting elements in the stress-responsive promoters function for plant adaptation to environmental stresses. It is estimated

that the *Arabidopsis* and rice genomes each contains about 1300–1600 genes that encode transcription factors [5]. As a result of computational prediction and manual curation, a total of 2576 putative transcription factors were identified and classified into 64 families from *Populus* [6].

Many transcription factors involved in these stress-resistance pathways have been identified, among which the AP2/ERF are one of the most important factors. AP2/ERF genes act as nodes of regulatory network in plant response to many stresses, which are conserved in many plants [7]. Nanjo and colleagues found 13 candidate clones that contained the AP2/ERF domain from the poplar ESTs [8]. They further analyzed cDNAs encoding an AP2/ERF-domain transcription factor which is specific in poplar and plays an important role under stress [9]. Two other DREB-like genes from *Populus* were isolated by yeast one-hybrid system [10]. To determine whether the C-repeat binding factor (CBF) family of transcription factors contributing to this process in annual herbaceous species also functions in woody perennials, the changes in phenotype and transcript profile of transgenic *Populus* constitutively expressing CBF1 from *Arabidopsis* (AtCBF1) were investigated [11]. Two ERF-related transcription factors were expressed during the transition to dormancy in apical and axillary buds [12].

In this research, the establishment of an extensive picture of the AP2/ERF gene family in *Populus* was attempted. First, we finely describe the poplar AP2/ERF gene family. The data obtained revealed 200 distinct AP2/ERF gene sequences in *Populus trichocarpa*. Secondly, we completed the inventory of the *Populus* AP2/ERF gene

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family. Sequence analyses and phylogenetic studies revealed the structural diversity in *Populus* AP2/ERF proteins. We named the genes coding for AP2/ERF domain proteins by following a nomenclature stemming from the phylogenetic analysis. The resulting classification of groups and identification of putative functional motifs will be useful in studies on the biological functions of each gene in the *Populus* AP2/ERF families.

Materials and methods

Source of *P. trichocarpa* and *Arabidopsis thaliana* database. The *P. trichocarpa* genome DNA Database was downloaded from DOE Joint Genome Institute web site (<http://genome.jgi-psf.org/Poptr1/Poptr1.download.html>) [2]. Database of *A. thaliana* AP2/ERF family was downloaded from DATF web site (<http://datf.cbi.pku.edu.cn/>) [6].

Database search. Searching of *P. trichocarpa* genome database was performed for finding all members of the AP2/ERF family. The strategy to obtain every gene of the AP2/ERF family in a genome is the following. The amino acid sequence of the AP2/ERF domain from tobacco (*Nicotiana tabacum*) ERF2 was used as queries to search against the JGI (US Department Of Energy, Joint Genome Institute) (<http://genome.jgi-psf.org/Poptr1/Poptr1.download.html>) databases using the BLASTP program at the e-value of 1e-3 to avoid false positives. To increase the extent of the database search results, we also performed the position-specific iterated BLAST search against the *P. trichocarpa* genome database on the NCBI web site. We also performed the database searches using amino acid sequences of the AP2/ERF domain of some members of the *Arabidopsis* AP2/ERF family as a query sequence to confirm completion of the collection, such as using the soloist At4g13040 sequence.

Sequences analysis and construction of the phylogenetic tree. Open reading frames (ORFs) were performed with ORF Finder (Open Reading Frame Finder) in NCBI (<http://www.ncbi.nlm.nih.gov/gorf/gorf.html>). A multiple alignment analysis was performed with ClustalW (<http://bioinformatics.ubc.ca/resources/tools/clustalx>) [13]. Phylogenetic trees were constructed using the neighbor-joining (NJ) method and the pictures of phylogenetic trees were drawn with program MEGA4 (<http://www.megasoftware.net/mega.html>) [14].

Bioinformatics analysis. Bioinformatics of AP2/ERF family transcription factors from *P. trichocarpa* and *A. thaliana*, such as gene and deduced amino acid sequence, composition, physical and chemical characterization, conserved domain sequences, and function domain, were analyzed using Expert Protein Analysis System (ExPASy) proteomics server of the Swiss Institute of Bioinformatics (SIB) (<http://cn.expasy.org/>) [15]. Recombinant proteins, solubility was predicted using the statistical model assuming the protein is being overexpressed in *Escherichia coli* at web site of University of Oklahoma (<http://biotech.ou.edu/>) [16]. The folding states of AP2/ERF family transcription factors from *P. trichocarpa* and *A. thaliana* were predicted by FoldIndex program (<http://bioportal.weizmann.ac.il/fldbin/findex>) [17].

The location of AP2/ERF Genes on *P. trichocarpa* chromosomes. Information on chromosomal location was gathered from the *Populus* genome browser (http://genome.jgi-psf.org/Poptr1_1/Poptr1_1.home.html). Chromosomal locations were determined according to chromosomal information from JGI.

Results

Identification of the AP2/ERF family transcription factors in *P. trichocarpa*

The completed genomes of *P. trichocarpa* and *A. thaliana* provide ideal systems for trans-genomic comparison of AP2/ERF among

these plant species. In order to carry out the cross-species comparison of the AP2/ERFs, a comprehensive list of AP2/ERF encoded in the two plant species has been obtained by sequence search as described under Materials and methods section. A total of 200 full length AP2/ERF family genes were identified as possibly encoding AP2/ERF domain (Table 1). The individual genes are listed in Supplementary Tables 1–5. This super-family included 5 putative genes in RAV family, which were identified as encoding one AP2/ERF domain together with one B3 domain. Twenty-six genes were predicted to encode protein containing two AP2/ERF domains, which were classified into AP2 family. One hundred sixty-eight genes were predicted to encode proteins containing a single AP2/ERF domain. Those 168 genes could be further classified into two families on the basis of similarity of the amino acid sequences. Seventy-seven genes were identified as possibly encoding DREB family, and 91 genes were predicted to encode ERF family. The remaining gene, eugene3.00002518, which was obtained using a soloist of AP2/ERF transcript factor from *A. thaliana* At4g13040 as query sequence, includes an AP2/ERF-like domain sequence, but its homology appears quite low in comparison with the other AP2/ERF genes. Therefore, this gene was also designated as a soloist.

Sakuma et al. reported 145 genes that are classified as members of the AP2/ERF super-family [18]. Of these genes, 121 were classified as part of the ERF subfamily and DREB subfamily, 17 were classified as part of the AP2 family, six were classified as part of the RAV family, and one remaining gene, At4g13040, was identical to the soloist gene. The number genes of total AP2/ERF family genes from *P. trichocarpa* (200) was 1.379-fold than those genes from *A. thaliana* (145). The number genes of DREB subfamily from *P. trichocarpa* (77) were 1.375-fold than those genes from *A. thaliana* (56). The number genes of ERF subfamily from *P. trichocarpa* (91) were 1.400-fold than those genes from *A. thaliana* (65). The number genes of AP2 subfamily from *P. trichocarpa* (26) were 1.529-fold than those genes from *A. thaliana* (17). The rates of total gene and same subfamily genes were similar between *P. trichocarpa* and *A. thaliana* (Fig. 1). Moreover, those rates were also similar with protein-coding genes of *Populus*/*Arabidopsis*. *Populus* has more protein-coding genes than *Arabidopsis*, ranging on average from 1.4 to 1.6 putative *Populus* homologs for each *Arabidopsis* gene [2].

Table 1
Summary of the AP2/ERF family between *P. trichocarpa* and *A. thaliana*

Plant		<i>P. trichocarpa</i>		<i>A. thaliana</i>	
Classification	Group	Number	Percent	Number	Percent
DREB subfamily	A1	6	3.00	6	4.14
	A2	18	9.00	8	5.52
	A3	2	1.00	1	0.70
	A4	26	13.00	16	11.03
	A5	14	7.00	16	11.03
	A6	11	5.50	9	6.21
	Total	77	38.50	56	38.63
ERF subfamily	B1	19	9.50	15	10.34
	B2	6	3.00	5	3.44
	B3	35	17.5	18	12.41
	B4	7	3.50	7	4.83
	B5	8	4.00	8	5.52
	B6	16	8.00	12	8.28
	Total	91	45.50	65	44.82
Ap2 subfamily		26	13.00	17	11.71
RAV subfamily		5	2.50	6	4.14
Solosist		1	0.50	1	0.70
Total		200		145	

The number of AP2/ERF family from *A. thaliana* was according to Sakuma et al. [18].

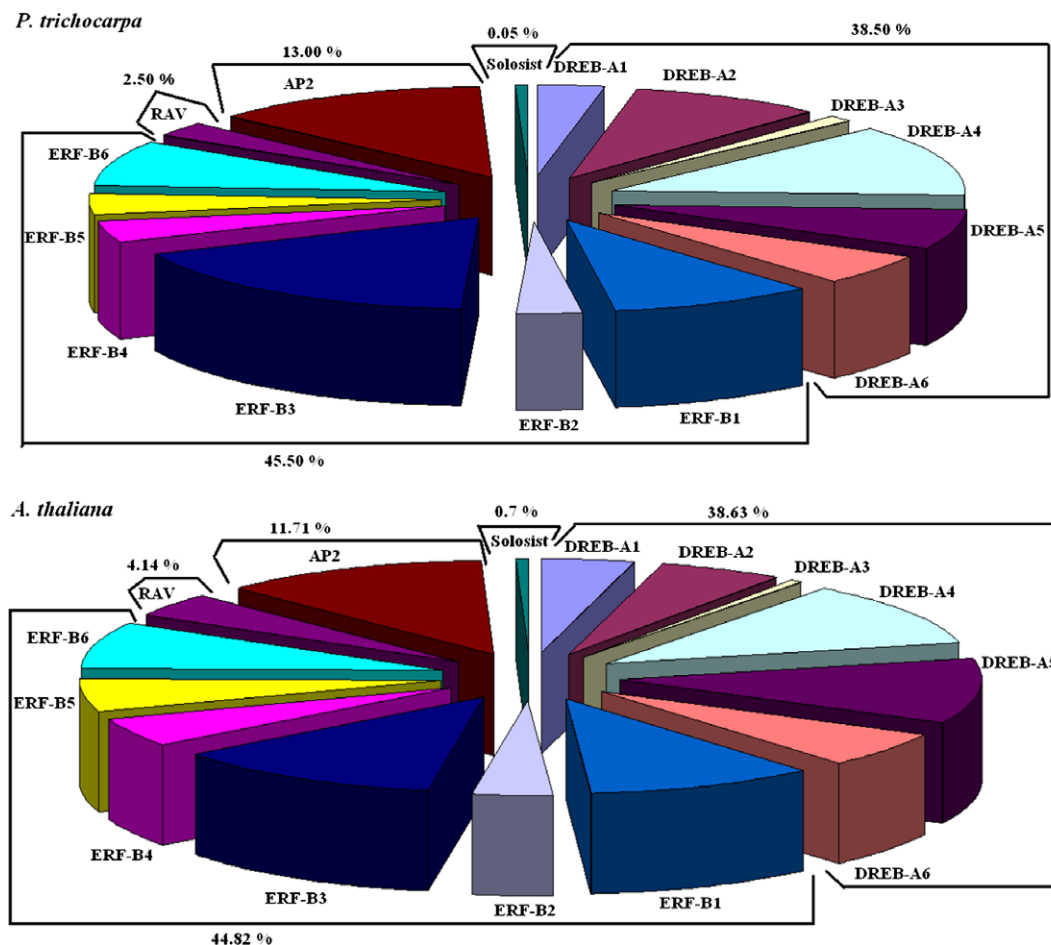


Fig. 1. The classification of AP2/ERF family genes between *P. trichocarpa* and *A. thaliana*. The size of each piece is proportional to the relative abundance to the AP2/ERF genes assigned to this group.

Phylogenetic relationships between the AP2/ERF family transcription factors in *P. trichocarpa*

Based on the alignment of the AP2/ERF domains of 200 AP2/ERF family genes of *P. trichocarpa* from this research and all AP2/ERF proteins from *A. thaliana*, three phylogenetic trees (full AP2/ERF family of *P. trichocarpa*, ERF subfamily of *P. trichocarpa* and *A. thaliana*, and DREB subfamily of *P. trichocarpa* and *A. thaliana*) were constructed. As shown in Fig. 2, 200 AP2/ERF family genes of *P. trichocarpa* were distributed in DREB, ERF, AP2, RAV family, and a soloist. As shown in Supplementary Fig. 1, the ERF family genes from *P. trichocarpa* were distributed in B1, B2, B3, B4, B5, and B6 subgroup. As shown in Supplementary Fig. 2, the DREB family genes from *P. trichocarpa* were classified into six subgroups, A1, A2, A3, A4, A5, and A6.

Sequence alignment and conserved motifs inside or outside of the AP2/ERF domain

To clarify the phylogenetic relationships of AP2/ERF proteins between *P. trichocarpa* in this research and *A. thaliana*, which have been deposited by others [18,19], multiple alignment analyses were performed using amino acid sequences of the AP2/ERF domain. Sequence alignment showed that all of subgroups B1, B2, B3, B4, and B5 of the ERF family factors from *P. trichocarpa* and *A. thaliana* contain 4G, 6R, 10W, 15A, 16E, 18R, 25R, 27W, 28L, 29G, 37A, 41Y, 42D, and 50G. (The number was accorded with the AT1G03800). The factors of B6 subgroup have more

variation with B1–B5 subgroup (Supplementary Fig. 3). From the sequences of B1–B6 subgroup alignments of the full length of putative amino acid from *P. trichocarpa* and *A. thaliana*, there was more variation outside of the AP2/ERF domain (Supplementary Figs. 4–9). Sequence alignment showed that all the DREB family genes containing WLG motif (W27, L28, and G29), except PtDREB-A1-2 contain V28. All the DREB family genes contain E16, R18, P20, 40A, and 42D in the DREB/CBF-DNA-binding domain (The number was accorded with the AT4G25470) (Supplementary Fig. 10). From the sequences of A1–A6 subgroup alignments of the full length of amino acid from *P. trichocarpa* and *A. thaliana*, there was more variation among those outside of the AP2/ERF domain (Supplementary Figs. 11–16). Twenty-six factors from *P. trichocarpa* were predicted to encode protein containing two AP2/ERF domains, which was assigned to the AP2 family (Supplementary Figs. 17 and 18). The PtRAV-1, PtRAV-2, PtRAV-3, PtRAV-4, and PtRAV-5 genes from *P. trichocarpa* were predicted to encode protein containing the AP2/ERF domain and the B3 domain. Those five factors were assigned to the RAV family (Supplementary Figs. 19 and 20).

Characterization of deduced amino acid sequences of AP2/ERF transcription factors from *P. trichocarpa*

Based on program of ExPASy and some other software, bioinformatics of AP2/ERF transcription factors from *P. trichocarpa* and *A. thaliana* were analyzed, such as number of amino acids, composition of deduced amino acid sequence, theoretical

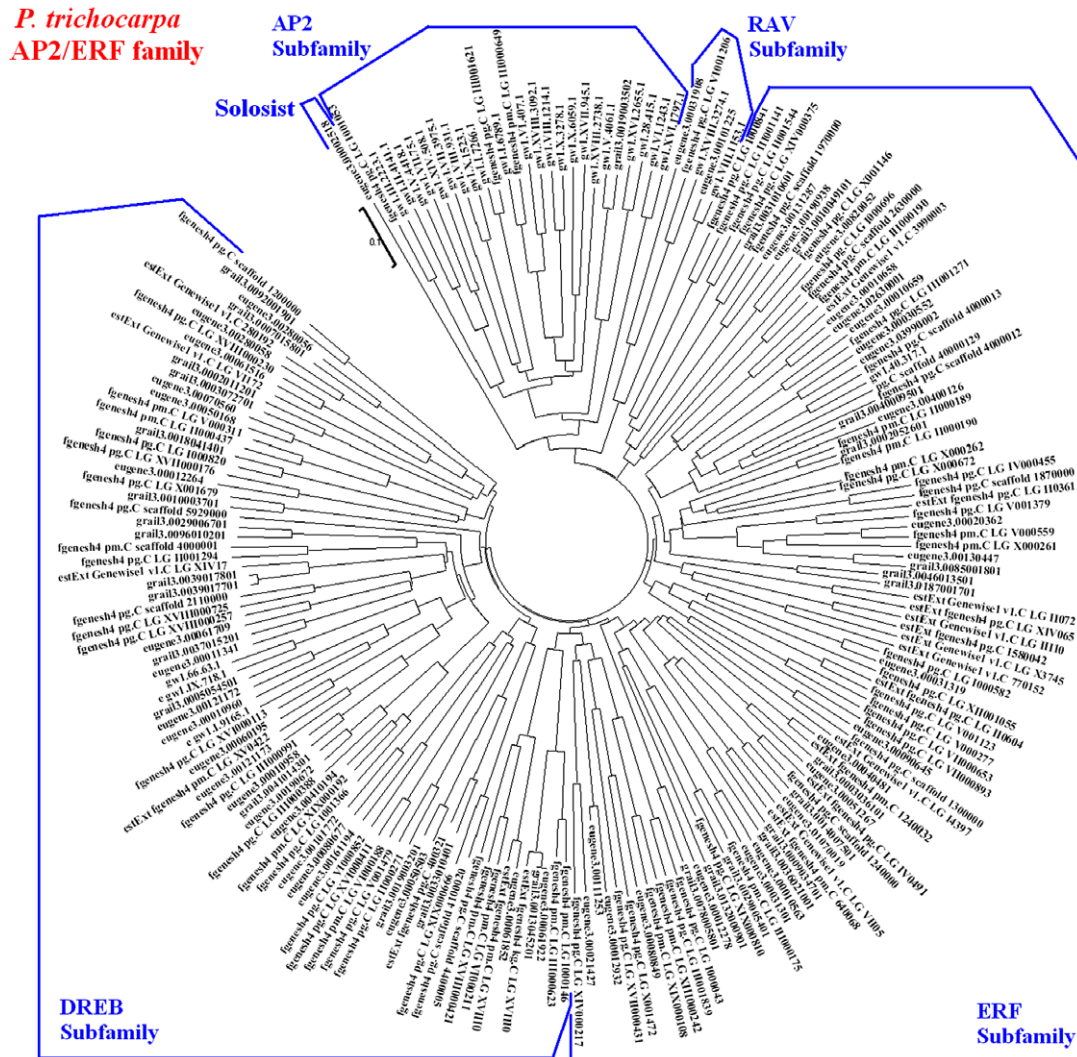


Fig. 2. Phylogenetic tree of all AP2/ERF family genes from *P. trichocarpa*.

Prediction of the AP2/ERF proteins' folding state from *P. trichocarpa*

The folding states of AP2/ERF family transcription factors from *P. trichocarpa* and *A. thaliana* were predicted by FoldIndex program (<http://biportal.weizmann.ac.il/fldbin/findex>). As shown in Supplementary Tables 20–33, the values' percentage of disordered residues of amino acid sequences of DREB-A1 subgroup transcription factors from *A. thaliana* was less than 70%. However, the values of DREB-A1 subgroup transcription factors from *P. trichocarpa* were over 70%. All of the DREB-A3 subgroup transcription factors from *A. thaliana* and *P. trichocarpa* were less than 50%. Among all proteins of AP2/ERF transcription factors from *A. thaliana* and *P. trichocarpa*, the highest of percentage of disordered was PtERFB2-4, which percentage of disordered residue was achieved 99.22%, and nearly completely disordered and contains large disordered regions in their native state. There only two amino acid residues (168K and 189Q) were folded. The lowest of percentage of disordered was RAV1 (from *A. thaliana* AT1G50680). The percentage of disordered residue was only 19.88%, less 20% unfolded (Fig. 3). Noticeably, the values' percentage of disordered residues of amino acid sequences among the PtERFB6-4 and PtERFB6-5 was 25.15% and 82.25%. Although the two proteins classified into an ERF subgroup, the folding states of amino acid were different over 3-fold (Fig. 3). **Supple-**

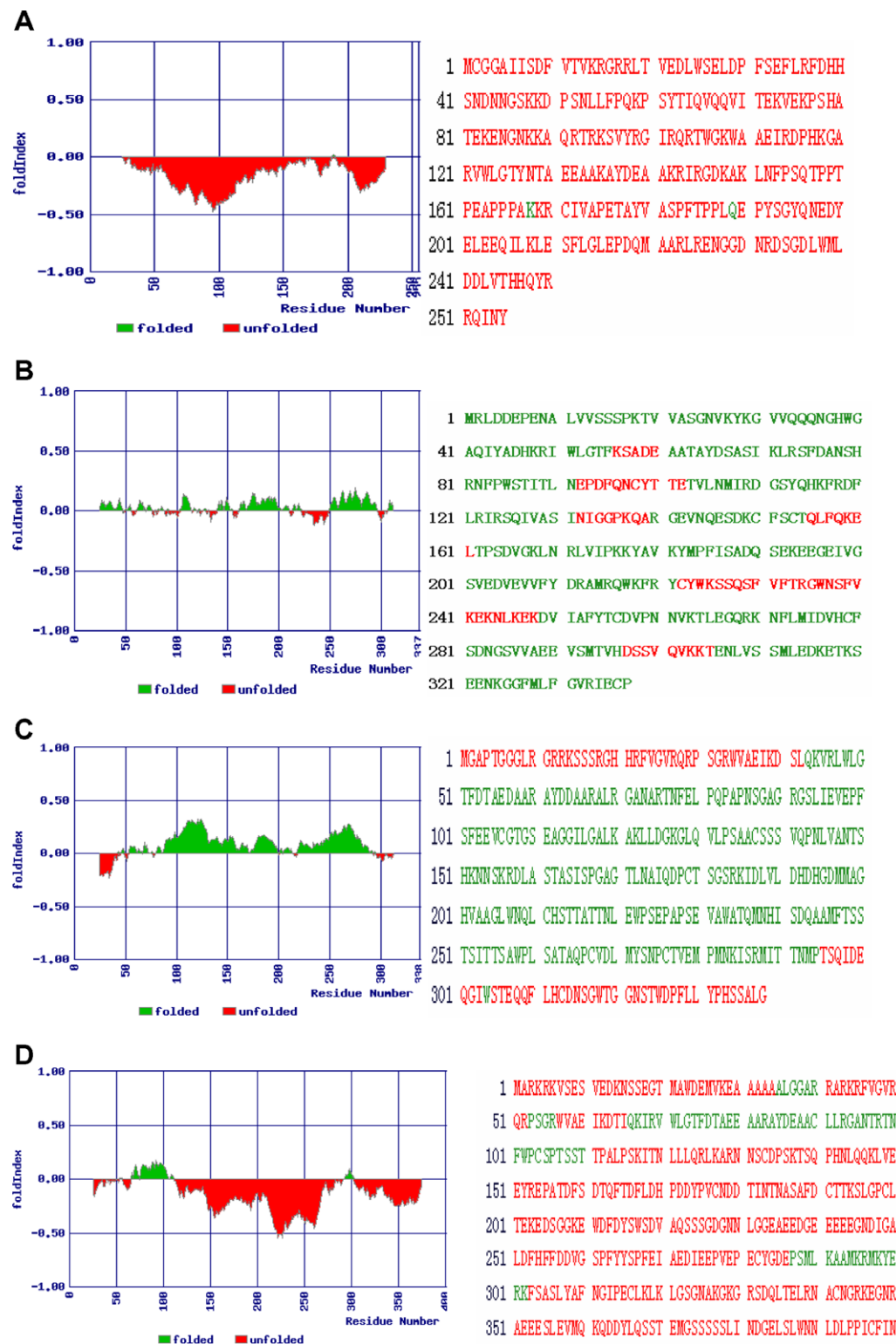


Fig. 3. Prediction of the folding state of PtERFB2-4, AtRAV1, PtERFB6-4, and PtERFB6-5. Positive and negative numbers represent ordered and nonordered protein, respectively. Amino acids suggested in (A) as being ordered and nonordered regions are shown in red and green characters, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

mentary Tables 20–33 showed prediction of the folding state of amino acid sequences (number disordered regions, longest disordered region, number disordered residues, and percentage of disordered residues) of all of the AP2/ERF transcription factors from *A. thaliana* and *P. trichocarpa*.

The locations of the AP2/ERF family genes on the *P. trichocarpa* chromosomes

The *Populus* genome has 200 AP2/ERF family genes. *In silico* mapping of the gene loci showed that 162 AP2/ERF family genes of

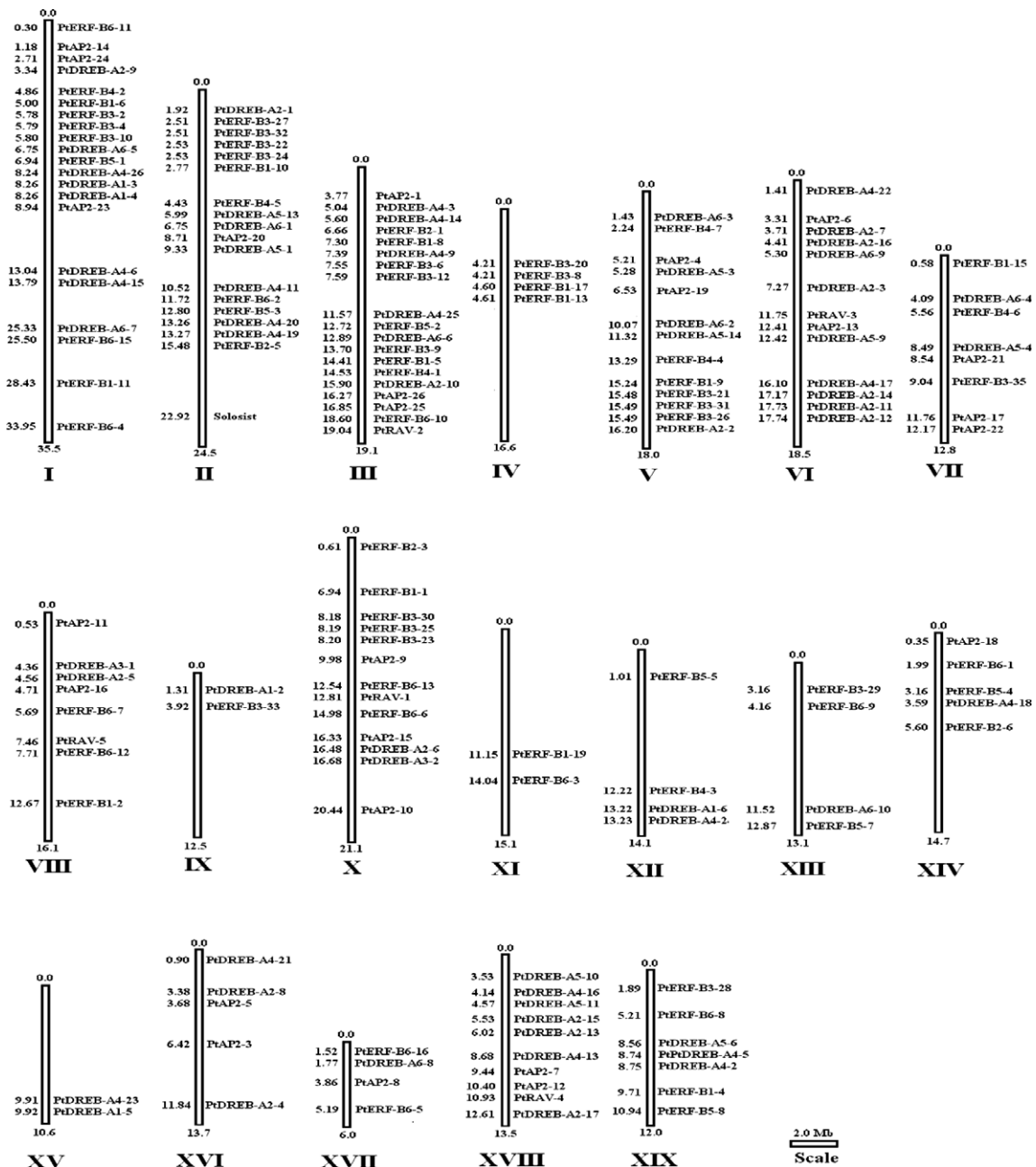


Fig. 4. Positions of AP2/ERF family genes on the *P. trichocarpa* chromosomes. Scale represents a 2.0 Mb chromosomal distance. Chromosome numbers (linkage group number I–XIX) and sizes (Mb) are indicated at the bottom end of each chromosome. Those on sequence scaffolds are not shown here. The locations are approximate and not proportional to their real physical distance.

Populus were localized on 19 linkage groups (Fig. 4) and the remaining 38 genes were localized on Scaffolds (Supplementary Fig. 21). Supplementary Fig. 22 depicts the distribution of AP2/ERF family genes in the *Populus* genome. Like in *Arabidopsis* and rice, poplar AP2/ERF genes seemed to be distributed among all the linkage groups. There were ten or more AP2/ERF family genes that were located on 7 chromosomes. There were five or less AP2/ERF family genes that were located on 9 chromosomes. Among them, 21 (10.5%) genes were localized on chromosome I, 18 genes (9%) on chromosome II, 19 genes (9.5%) on chromosome III, and 2 genes (1%) on chromosomes IX, XI, and XV (Fig. 4). We found that many AP2/ERF family genes were cluster localized on the linkage groups. Such as, PtERF-B3-2, PtERF-B3-4, and PtERF-B3-10 were cluster localized on ~20 kb segment (LG I), PtDREB-A4-26, PtDREB-A1-3,

and PtDREB-A1-4 were also cluster localized on another ~20 kb segment (LG I). PtERF-B3-21, PtERF-B3-31, and PtERF-B3-26 were cluster localized on ~10 kb segment (LG V). One segment on the linkage group VI (LG VI) was particularly interesting. This segment of <5 kb contained two AP2/ERF family genes that belong to the same DREB-A2 subgroup (PtDREB-A2-11 and PtDREB-A2-12). Furthermore, PtDREB-A2-11 and PtDREB-A2-12 have 100% identity protein sequence.

Discussion

Thousands of genes are involved in stress response and/or stress tolerance, according to recent developments in structural and functional genomics in some herbaceous model plant species such as

Arabidopsis and rice. Genes in the AP2/ERF family encode transcription regulators with a variety of functions involved in the developmental and physiological processes in plants. Isolation and identification of the functional genes and transcription factor genes from poplars were becoming one of the important researches, which could help us to make clear the molecular genetic basis for the poplar genetic improvement and also provide the functional gene resources for transgenic research, which also help us make clear the network of genes that control *Populus* development.

Poplar species are the important tree species for forestation and virescence, as well as for a large variety of wood-based products, also shape the ecology and productivity of the whole world. *Populus* has become the leading model organism in many laboratories to study the physiology, ecology, and genetics of forest trees for many reasons, such as rapid growth, diploid inheritance, and facile clone propagation. Another most important advantage of the poplars is that they have a small genome size, which is quite similar in size to that of rice and about four times larger than that of *Arabidopsis*, organized in 19 chromosomes [20]. Poplar genome is the first to be sequenced among woody plants. The nucleotide sequence of the entire genome of black cottonwood was released of version 1.1 of by the DOE JGI. The sequence will allow the development of a strategy for improving environmental stress tolerance in forest trees [2]. Tuskan et al. reported that *Populus* has more protein-coding genes than *Arabidopsis*, ranging on average from 1.4 to 1.6 putative *Populus* homologs for each *Arabidopsis* gene. In this research, 200 AP2/ERF family genes were *in silico* cloning from *Populus* [2]. Of these genes, the number genes of total AP2/ERF family genes, DREB subfamily, ERF subfamily, and AP2 subfamily from *P. trichocarpa* were also about 1.4–1.6-fold than those genes from *A. thaliana*. Furthermore, the AP2/ERF family genes were very similar from woody (poplar) and herbaceous (*Arabidopsis*) plants. We think that the function and network, those AP2/ERF family genes were also similar between those plants. The resulting classification of groups, identification of putative function motifs of and analysis of bioinformatics properties will be useful in studies on the biological functions of each gene in the AP2/ERF families of *Populus*. In this work, many “hot” sites were localized on *Populus* chromosomes, where some AP2/ERF family genes were clustered. Few AP2/ERF transcription factors were localized on some chromosomes (LG IX, LGIX, and LG XV), while many AP2/ERF transcription factors were localized on some other chromosomes (LG I, LG II, LG III, and LG X). Further studies maybe provide the functional gene resources for transgenic research, which also help us make clear the network of genes of *Populus*.

Primary protein structure analysis, secondary protein structure prediction, tertiary protein structure model, and natively unfolded protein study were important research fields of life science. To be folded or to be unfolded was a field of central dogma, which has not been studied clearly until now. It is now possible to predict extended disordered regions in genome sequences with a high degree of accuracy. The folding state of abscisic acid stress ripening 1 (ASR1) from Tomato (*Solanum lycopersicum*) was predicted by the FoldIndex program. ASR1 protein is mostly unfolded under physiological conditions [21]. The highest of percentage of disordered residue was achieved 99.22% (PtERFB2-4). The protein was completely disordered and contains large disordered regions in their native state. Although the two proteins classified into an ERF subgroup, the folding states of amino acid were different over 3-fold (PtERFB6-4, 25.15%; PtERFB6-5, 82.25%). Those differences of folding state of AP2/ERF transcription factors from *A. thaliana* and *P. trichocarpa* are research fields which are worth studying in the future.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.bbrc.2008.04.087.

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